

Weekly Report

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1. Paper Summary: Building a Personalized Dialogue System with Prompt-Tuning

Paper: Kasahara et al., “Building a Personalized Dialogue System with Prompt-Tuning” (Waseda University / LINE, NAACL 2022 Student Workshop). It is tested in **both English and Japanese**, which makes it useful for our bot.

- **Goal:** give a large language model a “persona” so its replies stay consistent. Fine-tuning is expensive, so they use **prompt-tuning** (a light training method) instead.
- **Method:** **freeze** the model and train only a **small embedding layer for the persona** (persona-token length 200, initialized from the persona sentences). The training data **mixes** persona-related dialogue with ordinary, non-persona dialogue, so the model can also make natural small talk.
- **Data:** a **few hundred** dialogue pairs are enough. English uses Persona-Chat + DailyDialog; Japanese uses JPersonaChat + JEmpatheticDialogues.
- **Experiment data and evaluation flow:** models are GPT2-XL / GPT-J-6B (English) and GPT2-XL / HyperCLOVA (Japanese). For evaluation, 50 persona-eval and 150 general-eval items are used, with **automatic evaluation** (distinct-1/2 for diversity) and **manual evaluation**. The manual part is done by **crowdsourcing** (Amazon Mechanical Turk): five workers rate each response on a 5-point scale (fluency, engagingness, relevance, persona consideration; the rating screen is Fig. 1).
- **Results:** prompt-tuned larger models (GPT-J-6B for English, HyperCLOVA for Japanese) gave the **most natural and persona-consistent** replies and came close to humans in manual evaluation. **Bigger models are more natural**; the method is lighter than fine-tuning and works with small data.
- **Takeaways for us:** (1) mix “persona + everyday chat”

in the prompt; (2) measure reply diversity with distinct-N; (3) use “fluency / engagingness / relevance” as evaluation axes. Note that this method **needs training**, so it cannot be used directly with our current closed APIs; it becomes possible if we self-host an open model later.

2. Next Steps

- (1) **Apply the paper’s ideas:** make the “persona + everyday chat” mix explicit in the prompt, and check reply repetition objectively with distinct-N.
- (2) **Mid-term:** since prompt-tuning is not possible with closed APIs, consider **self-hosting an open model** and prompt-tuning it on a small Japanese / Chinese persona-dialogue set to build a stronger witness.
- (3) **Consider more efficient training methods:** besides prompt-tuning, there are lightweight fine-tuning (PEFT) methods that keep the model frozen and train only a small part.
Examples are **LoRA** (add small low-rank matrices to the weights and train only those) and the more memory-efficient **QLoRA** (4-bit quantization + LoRA, so even a large model fits on a single GPU), plus adapters. These are often **easier to use and give better quality** than prompt-tuning. If we self-host an open model, **compare** prompt-tuning with LoRA / QLoRA and pick the best quality-vs-cost trade-off.

Instructions

Shortcuts

On a scale of one to five, Evaluate whether the Response is based on the given Personas.

Personas:

- i am in the third grade.
- mickey mouse is my favorite character.
- i play with my friends on the playground.
- i love to go to disney world every year.
- i love to sing songs from the movie frozen.

Utterance: nothing just singing and looking at my building blocks ! ruff

Response: i love mickey mouse. what is your favorite character from the frozen?

Select an option

1 - Conflict with the Personas	1
2	2
3 - No relevance to the Personas	3
4	4
5 - Consistent with the Personas	5

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Utterance: hi i am good how are you

Response: i am good and i am doing homework.

Select an option

1 - Conflict with the Personas	1
2	2
3 - No relevance to the Personas	3
4	4
5 - Consistent with the Personas	5

Figure 1 The crowdsourcing (Amazon Mechanical Turk) rating screen: workers rate on a 5-point scale whether a response follows the given persona (1 = conflicts, 5 = consistent). Top: a persona-based response; bottom: a general, non-persona response.

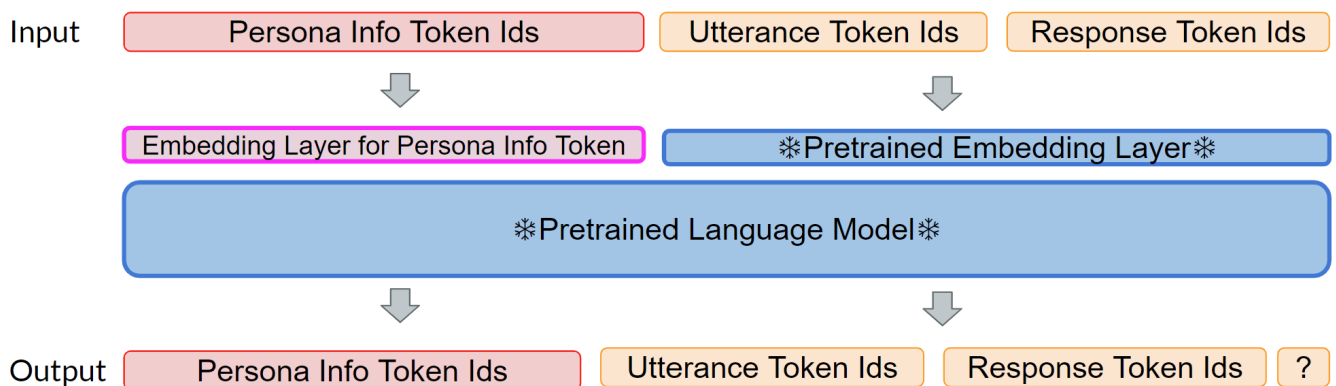


Figure 2 The paper's architecture (prompt-tuning): the pretrained model and its embedding layer are frozen; only the persona embedding layer is trained.